PROJECT REPORT

EV Market Analytics Dashboard

Interactive Dashboard for Electric Vehicle Sales, Manufacturer and Battery Technology Insights

Team ID	XXXXXX
Project Name	EV Market Analytics Dashboard
Date	15 March 2026
Organisation	SmartBridge / SkillWallet

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1. INTRODUCTION

1.1 Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market, driven by environmental awareness, government incentives and advances in battery technology. The sector produces large amounts of data on vehicle sales, pricing, battery specifications, manufacturing locations, customer ratings and sustainability metrics.

This project delivers an interactive analytics dashboard that turns this data into clear, decision-ready insights.

The dashboard has three pages: an Executive Overview, a Manufacturer and Market Share page, and a Battery and Technology page. Shared filters for year, vehicle category, manufacturer, battery type and manufacturing country apply across every page. It is designed for three user personas: Emily Johnson (EV Market Strategy Director), David Chen (Automotive Business Intelligence Manager) and Dr. Sarah Williams (Sustainable Mobility Research Analyst).

1.2 Purpose

The purpose of the project is to help manufacturers, consultants, investors and researchers:

- Identify which vehicle categories contribute most to annual sales and which segments are growing fastest.
- Compare manufacturers by market share, sales performance and customer satisfaction.
- Understand battery technology adoption (LFP, NMC, NCA, Sodium-Ion, Solid-State) and compare driving range across categories.
- Analyse global manufacturing distribution by country.
- Support strategic planning, product positioning, market expansion and investment decisions.

2. IDEATION PHASE

2.1 Problem Statement

EV market data is large, spread across many attributes and difficult to interpret in raw tables.

Decision-makers cannot quickly see which categories, manufacturers and battery technologies are driving growth, or how customer satisfaction relates to performance. A single, interactive and easy-to-read dashboard is needed so that different stakeholders can explore the same data from their own perspective.

Problem Statement	I am	I am trying to	But	Because	Which makes me feel
PS-1	Emily Johnson, Market Strategy Director	Identify top categories, leading manufacturers and high-growth segments	Data is scattered and hard to compare	There is no single executive view	Uncertain about portfolio decisions
PS-2	David Chen, BI Manager	Advise clients on market entry and positioning	Market share and country data are not easy to compare	Reports are static and manual	Slow and less confident in recommendations
PS-3	Dr. Sarah Williams, Research Analyst	Compare battery technologies and range improvements	Technology trends are buried in raw data	There is no visual comparison across battery types	Limited in producing evidence for policy

2.2 Empathy Map Canvas

The empathy map below summarises the primary user group (analysts and decision-makers in the EV industry).

SAYS	THINKS
"Which category sells the most?" "Which battery type is winning?" "I need this in one view."	Data should be quick to trust and easy to explain to leadership. Insights must lead to a decision.
DOES	FEELS
Compares sales, range and ratings in spreadsheets, builds manual reports and presents findings to stakeholders.	Overwhelmed by raw data, pressured by tight timelines, and wants confidence in the numbers.
PAINS	**GAINS**
Scattered data sources, manual calculations, no way to filter by region or battery, difficulty spotting trends.	A single interactive view, quick answers by filtering, clear insight into growth segments, and data-backed recommendations.

2.3 Brainstorming

The team brainstormed what each persona would need and grouped ideas by theme. Ideas were then prioritised by value to users and ease of implementation.

Theme	Ideas generated	Decision
Sales	Sales by category, yearly trend, growth rate, top categories	Included
Manufacturers	Market share donut, top 10 manufacturers, sales vs rating scatter	Included
Geography	Manufacturing country map, sales by country	Included
Battery & Technology	Battery type distribution, range by battery, capacity vs range, adoption over time	Included
Interactivity	Global filters (year, category, manufacturer, battery type, country)	Included
Advanced	Sales forecasting, live data refresh, charging infrastructure view	Moved to future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Awareness	Exploration	Analysis	Decision	Follow-up
User action	Needs EV market insight	Opens dashboard and applies filters	Compares categories, manufacturers and batteries	Draws conclusions and picks a strategy	Shares findings and revisits with new filters
Touchpoint	Business question / client brief	Overview page	Manufacturer and Battery pages	Key insight visuals	Exported views / presentation
Feeling	Curious, uncertain	Interested	Engaged, confident	Satisfied	Trusting
Opportunity	Clear landing KPIs	Simple, visible filters	Consistent colours and tooltips	Highlight top insights	Easy sharing and refresh

3.2 Solution Requirement

ID	Type	Requirement
FR-1	Functional	Load and clean the EV dataset (sales, price, battery type, capacity, range, country, rating).

FR-2	Functional	Display KPI cards: total sales, average range, battery capacity, rating and number of manufacturers.
FR-3	Functional	Show sales by vehicle category, yearly trend and growth rate.
ID	**Type**	**Requirement**
FR-4	Functional	Show manufacturer market share, top manufacturers and rating comparison.
FR-5	Functional	Show battery type distribution, range by battery and capacity vs range.
FR-6	Functional	Provide global filters that update every visual.
NFR-1	Non-functional	Dashboard should respond to filters within a couple of seconds.
NFR-2	Non-functional	Consistent, readable design with clear labels and colours.
NFR-3	Non-functional	Data accuracy and consistent definitions of each metric.
NFR-4	Non-functional	Accessible on common screen sizes and easy to share.

3.3 Data Flow Diagram

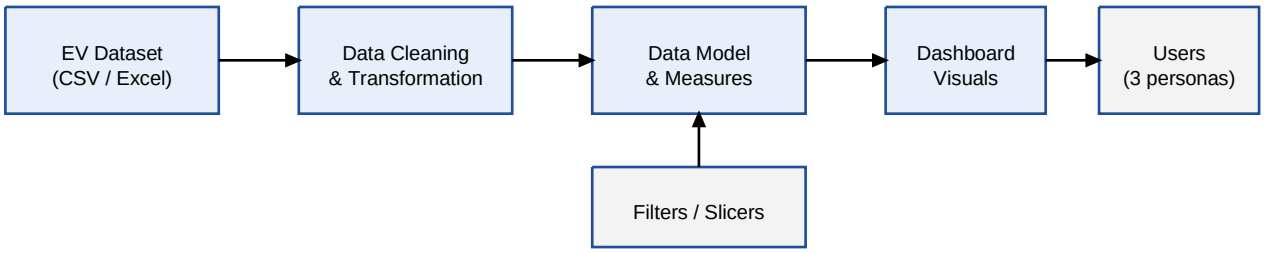


Figure 1: Data flow from raw dataset to user insights.

3.4 Technology Stack

Layer	Technology
Data source	EV dataset in CSV / Excel format
Data preparation	Power Query / Excel for cleaning and transformation
Analytics & visualisation	Power BI / Tableau (interactive dashboard, DAX measures, slicers)
Version control & sharing	GitHub and Google Drive
Collaboration	Google Meet, WhatsApp group

4. PROJECT DESIGN

4.1 Problem Solution Fit

Customer problem	Solution fit
Cannot see which categories drive sales	Sales by category with trend and growth view

Hard to compare manufacturers	Market share and top-manufacturer visuals with ratings
Unclear which battery technology leads	Battery distribution, range and capacity comparison
Needs different views for different roles	Three pages with shared filters, one per persona

4.2 Proposed Solution

Aspect	Description
Problem being solved	Turning large EV data into clear, comparable, decision-ready insights.
Idea	A three-page interactive dashboard with shared filters.
Aspect	**Description**
Novelty	One dashboard serving strategy, consulting and research personas with consistent metrics.
Social impact	Supports evidence-based policy and faster, informed EV adoption.
Business model	Usable by consulting firms, manufacturers and research bodies as a reusable analytics product.
Scalability	Can extend to live data, forecasting and larger regional datasets.

4.3 Solution Architecture

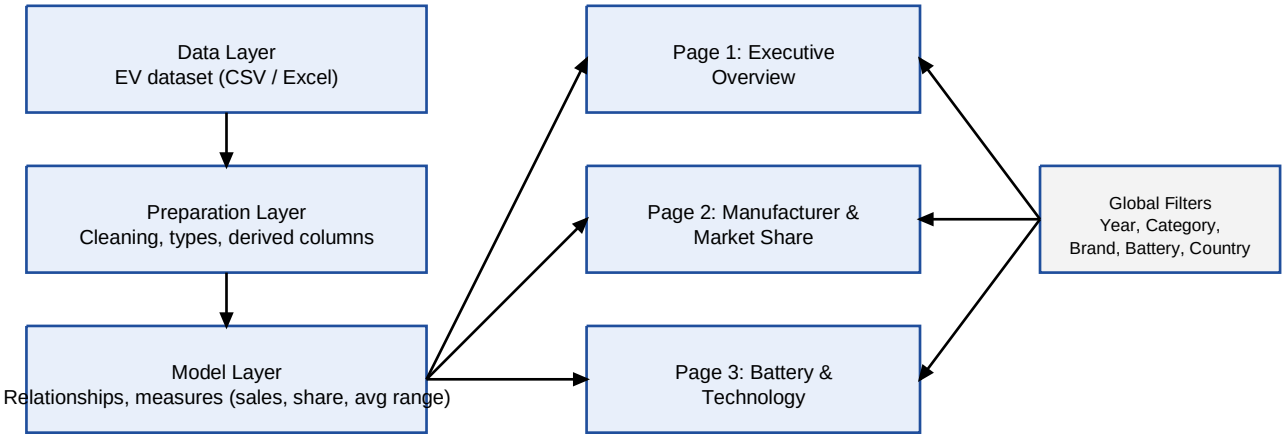


Figure 2: Solution architecture.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Task	Description	Duration	Status
Sprint 1	Requirement gathering	Study scenarios and define KPIs for the three personas	3 days	Completed
Sprint 1	Data preparation	Collect, clean and validate the EV dataset	4 days	Completed
Sprint 2	Data modelling	Create measures and calculated fields	3 days	Completed
Sprint 2	Executive Overview page	KPIs, sales by category, trends and ratings	4 days	Completed

Sprint 3	Manufacturer page	Market share, top manufacturers and country analysis	4 days	Completed
Sprint 3	Battery & Technology page	Battery distribution, range and capacity comparison	4 days	Completed
Sprint 4	Filters & testing	Global filters, performance and functional testing	3 days	Completed
Sprint 4	Documentation & demo	Report, demo plan and rehearsal	3 days	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

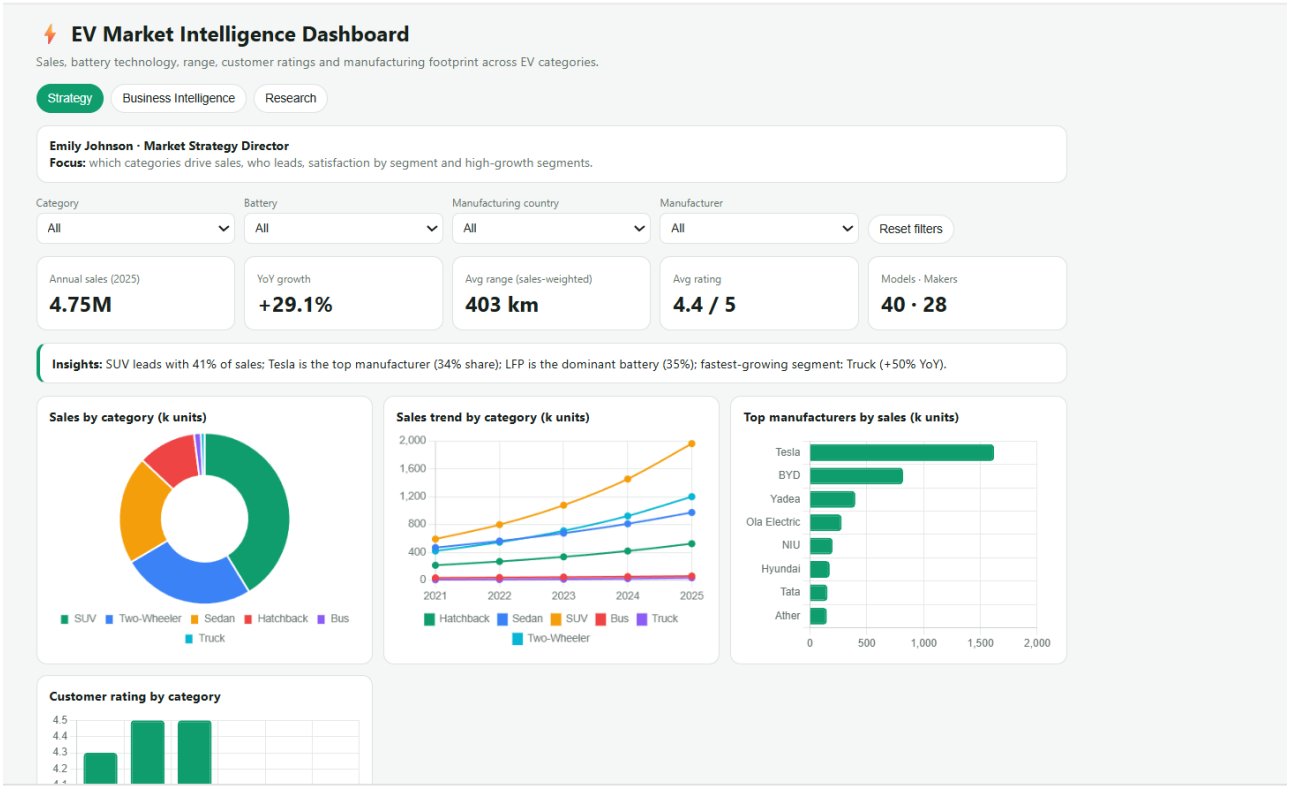
6.1 Performance Testing

The following test cases were prepared to confirm that the dashboard behaves correctly and performs well.

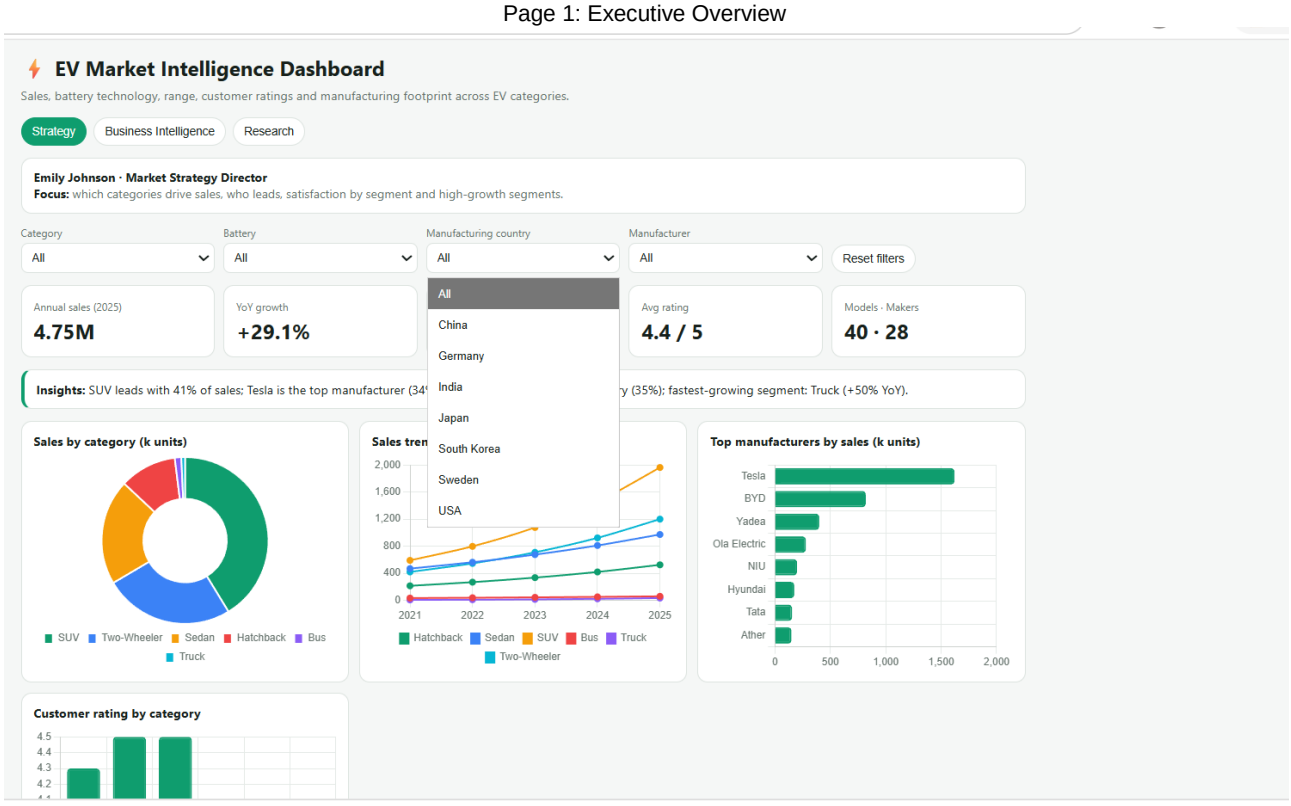
S.No	Test case	Expected result	Result
1	Apply the year filter	All visuals update to the selected year	Pass
2	Apply vehicle category filter	KPIs, charts and tables reflect the category	Pass
3	Select a manufacturer	Market share and ratings update correctly	Pass
4	Select a battery type	Range and capacity visuals update correctly	Pass
5	Check KPI totals against source data	Totals match the dataset	Pass
6	Combine multiple filters	Visuals show consistent filtered data	Pass
7	Dashboard load time	Loads and refreshes within a few seconds	Pass
8	Navigate between pages	Filters and layout stay consistent	Pass

7. RESULTS

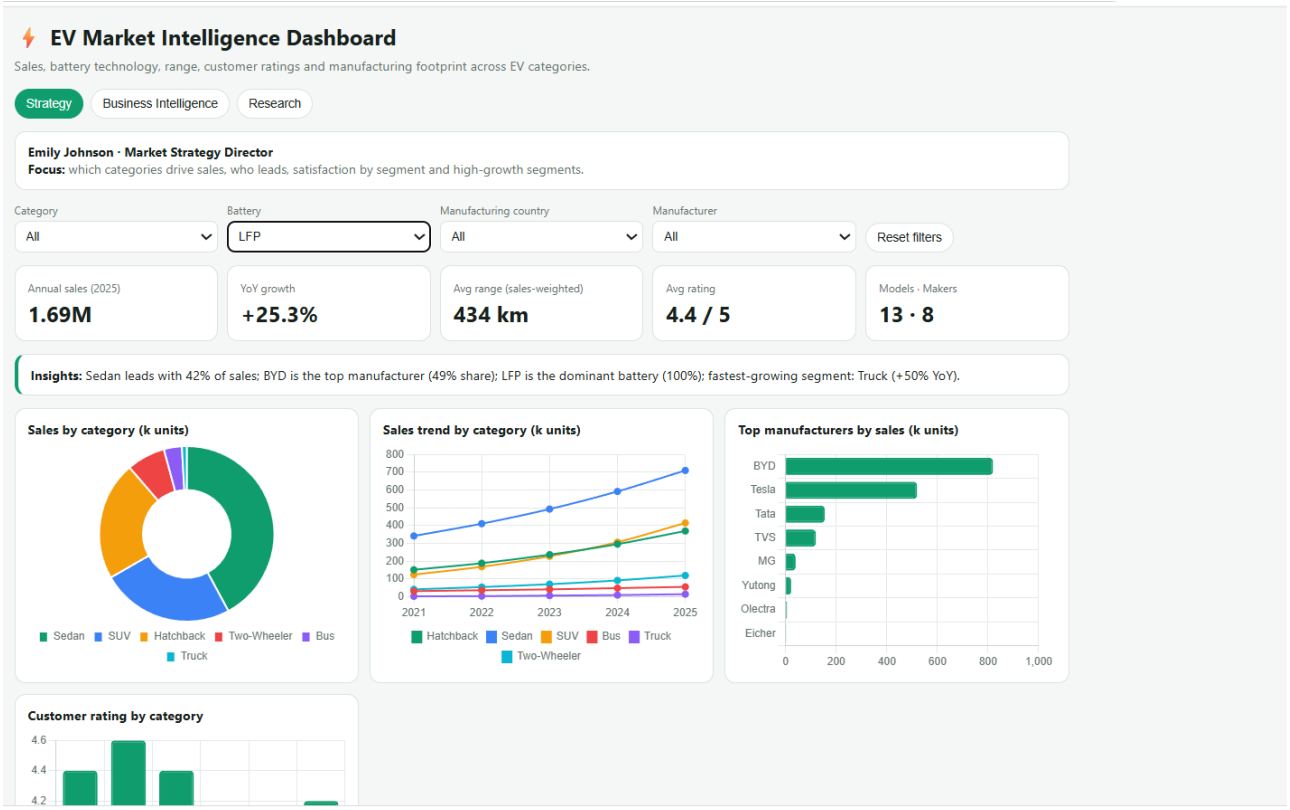
7.1 Output Screenshots



Page 1: Executive Overview



Page 2: Manufacturer and Market Share



Page 3: Battery and Technology

Key insights the dashboard is designed to show

- Which vehicle categories contribute most to annual sales and which are growing fastest.
- Which manufacturers hold the largest market share and how that compares with customer ratings.
- Which battery technologies dominate and how driving range varies by battery type and category.
- Which countries lead EV manufacturing.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Single interactive view for strategy, consulting and research needs	Uses a static dataset that must be refreshed manually
Global filters allow fast, flexible exploration	No forecasting or predictive analysis yet
Clear visuals make insights easy to share with leadership	Limited to the attributes present in the dataset
Consistent metric definitions across pages	Needs a BI tool licence or environment to view and edit

9. CONCLUSION

The EV Market Analytics Dashboard brings sales, manufacturer, geography and battery technology data into one interactive tool. It meets the needs of all three personas by showing high-growth segments, competitive positioning and technology trends through shared filters and consistent visuals. The project shows how data visualisation can support faster, better-informed decisions in the EV industry and contribute to the wider move toward sustainable mobility.

10. FUTURE SCOPE

- Automated data refresh from live sources.
- Sales and battery-adoption forecasting.
- Charging infrastructure and sustainability metrics.
- Regional and country-level price and incentive comparison.
- Cloud deployment with role-based access and mobile-friendly views.

11. APPENDIX

Item	Details
Source Code (if any)	DAX measures / calculated fields: [add here or attach]
Dataset Link	[add dataset link]
GitHub & Project Demo Link	[add GitHub repository link] | [add demo video / dashboard link]